

Volume III · Chapter 16 — Deployment, FHIR Integration & Compliance · Theory

Source: split from `med-tech-curriculum-V1.md` (V1). From this split onward this chapter is maintained **independently** here. See `Volume-III-Chapter-16-context.md` for the AI interaction log.

Prerequisites: Ch 12, Ch 15, Ch 3.

Learning outcomes — the student can: deploy models locally (Ollama/Open WebUI); build an inference API; integrate via FHIR (read/write, SMART-on-FHIR); monitor performance/drift; navigate the regulatory/compliance landscape.

Topics (theory) — 10 h:

#	Topic	h
16.1	Deployment options: local serving, Ollama, Open WebUI, APIs	2
16.2	Inference serving: latency, batching, quantization, scaling	2
16.3	FHIR integration & SMART-on-FHIR app model (read/write)	2
16.4	Monitoring, logging, drift & feedback loops	2
16.5	Regulatory & compliance landscape (SaMD, FDA/CE, HIPAA/GDPR at deployment)	2

Datasets/tools: Ollama, Open WebUI, FastAPI, HAPI FHIR/Synthea; monitoring tools.

Assessment: deployed FHIR-integrated service (**60%**); monitoring setup (**20%**); quiz (**20%**).

Key decisions: local vs. cloud; API design; regulatory pathway; PHI handling at inference.

References: plan §13 → ...*local inference*; *Interoperability standards*; *Clinical data, code sets & privacy*. **Hours:** Theory **10** + Lab **16** = **26**.